Excel Assignment 2: Data Cleaning & Transformation

Project Summary & Executive Deliverable Report

Project Description: Real-world retail and product catalogs routinely arrive with missing attributes, casing anomalies, structural concatenation, duplicate records, and unstructured text codes. This project implements an end-to-end data preprocessing workflow on the **Product Dataset** using Microsoft Excel. The pipeline resolves data quality defects across missing categories and prices, standardizes text casing and spelling errors, removes redundant rows, splits atomic identifiers into meaningful dimensions (Manufacturing Date and Country Code), merges brand attributes, and applies business-grade visual formatting.

Key Project Metrics & Pipeline Summary

Metric / Stage	Before Preprocessing	After Preprocessing	Core Operation Applied
Total Rows	34 Records	31 Unique Records	Deduplication across all columns (3 duplicate rows removed)
Category Typos	5 rows ('Electroni')	0 errors ('Electronics')	Find & Replace: 'Electroni' → 'Electronics'
Product Name Casing	5 lowercase instances	100% Proper Title Case	Text Standardization: PROPER() casing standardization
Missing Categories	4 null values (11.8%)	0 nulls (100% complete)	Contextual mapping: Sneakers → Fashion, Coffee Maker → Kitchen, etc.
Missing Prices	3 null values (8.8%)	0 nulls (100% complete)	Benchmark imputation: Sunglasses ($150), Headphones ($250), Tent ($130)
Structural Columns	Product ID (composite)	Manufacturing Date + Country Code	Delimiter splitting ('-') into DD-MM-YYYY Date and ISO Country Code
Brand & Name	2 separate columns	Unified 'Product Brand'	CONCATENATE / '&' Operator merging Brand and Product
Formatting & UX	Raw numbers, no rules	Currency, Dates & Highlights	Currency ($#,##0.00), Date (DD-MM-YYYY), Green fill for Electronics

Evaluation Rubric Alignment & Methodology

1. **Missing Value Handling:** Imputed missing categories using exact product-to-category associations within the dataset. Missing prices were imputed using matched product benchmarks and category medians without introducing distribution distortion.
2. **Text Standardization & Typo Correction:** Eliminated grammatical casing variations (e.g., 'laptop' → 'Laptop') and corrected systematic truncation typos ('Electroni' → 'Electronics') to ensure accurate filtering and pivot table aggregation.
3. **Duplicate Removal:** Executed full-row deduplication, eliminating 3 exact redundant rows (HP Laptop, Bose Headphones, Samsonite Laptop Bag), establishing 100% referential integrity.
4. **Data Transformation (Splitting & Merging):** Extracted manufacturing timestamps and country origin codes from the composite Product ID string, while concatenating brand and product into an intuitive commercial display label.
5. **Number Formatting & Conditional Visualization:** Formatted monetary fields with currency formatting, standardized manufacturing dates to DD-MM-YYYY, implemented soft-green highlight rules for 'Electronics', and added price data bars for instant visual triage